Neuron Model Options for the Distribution of Oscillators
C. elegans Locomotion Circuit

Distribution of Oscillators Project

March 23, 2026

Abstract

The current piecewise-linear (Yuval) neuron model exhibits intrinsic bistability that creates network-level locking and prevents reliable oscillations. This document surveys three alternative models—FitzHugh–Nagumo, Izhikevich, and a graded potential model—evaluating each against the project's core requirements: few free parameters, no pathological bistability, and biological appropriateness for *C. elegans* locomotion.

1 Motivation

The current model uses a piecewise-linear approximation to the membrane voltage dynamics with three key voltage levels: resting potential L, threshold T, and plateau H. This produces a neuron with *two stable fixed points*—rest ($V = L$) and plateau ($V = H$)—separated by an unstable threshold at T.

In isolation this bistability is useful: it distinguishes oscillator cells (driven above T) from passive cells (below T), which is central to the Distribution of Oscillators framework. However, at the network level it creates two serious problems.

1. **Plateau locking.** Any excitatory connection strong enough to push a target neuron above T locks it at the plateau indefinitely. The neuron has no intrinsic restoring force back below T once it crosses that boundary.
2. **Cascade freezing.** Because excitatory neurons in the six-segment network (types 1–3) receive no inhibitory synaptic input, a single locking event can propagate through the excitatory subnetwork and freeze the entire circuit.

The fundamental requirement for a replacement model is therefore: a neuron that can oscillate or be driven to fire, but which *returns to rest naturally* when synaptic drive is removed, without requiring an external hyperpolarizing current.

2 Model Descriptions

2.1 FitzHugh–Nagumo (FHN)

$$\frac{dv}{dt} = v - \frac{v^3}{3} - w + I \tag{1}$$

$$\frac{dw}{dt} = \frac{1}{\tau}(v + a - bw) \tag{2}$$

Parameters: a, b, τ, I (4 total).

The variable v represents membrane voltage and w is a slower recovery variable analogous to potassium channel gating. The cubic nullcline of v produces a single stable fixed point for I below the bifurcation and a limit cycle for I above it. Crucially, the rest state is *globally attracting* when I is removed—there is no plateau to lock onto.

The oscillator/passive distinction maps cleanly onto I relative to the Hopf bifurcation point, preserving the conceptual structure of the OscillatorComb framework.

2.2 Izhikevich

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I \tag{3}$$

$$\frac{du}{dt} = a(bv - u) \tag{4}$$

with the reset rule: if $v \geq 30\,\mathrm{mV}$ then $v \leftarrow c$, $u \leftarrow u + d$.

Parameters: a, b, c, d (4 total, plus I).

The Izhikevich model is a quadratic integrate-and-fire neuron augmented with an adaptation variable u. The four parameters can be tuned to reproduce a wide variety of firing patterns (tonic spiking, bursting, chattering, adaptation) without changing the model structure. The reset rule replaces the plateau mechanism entirely, so there is no intrinsic bistability.

2.3 Graded Potential Model

$$\tau\frac{dV}{dt} = -V + \tanh\big(\beta(I_{\text{syn}} + I_{\text{app}})\big) \tag{5}$$

Parameters: τ, β (2 total per cell, plus I_{app}).

This is a rate-coded model in which V represents the mean firing rate or graded membrane potential rather than an instantaneous voltage. The sigmoidal transfer function tanh bounds the output and provides a smooth, globally stable response. There is no threshold crossing and no bistability by construction.

This model is the most commonly used in published *C. elegans* locomotion simulations [1, 2, 3] because of its biological basis: the majority of *C. elegans* neurons are thought to be **non-spiking**, communicating via graded potentials rather than action potentials [4].

3 Comparison

	Yuval PL	**FHN**	**Izhikevich**	**Graded**
Free parameters	8	4	4	2
Bistable	Yes	No	No	No
Intrinsic oscillations	Via SF	Yes	Yes	No (network only)
Plateau locking risk	High	None	None	None
OscillatorComb compatible	Yes	Yes	Yes	Requires redesign
Biophysical realism	Low	Low	Medium	Medium–High
C. elegans appropriate	Unclear	Unclear	Unclear	**Yes**
Computational cost	Low	Low	Low	Very low
Mathematical tractability	High	High	Medium	High

4 Tradeoffs

4.1 FitzHugh–Nagumo

FHN is the natural successor to the current model. It retains the two-variable structure (voltage + recovery, analogous to V + SF), has the same parameter count as the project requires, and its bifurcation structure is well-understood analytically. The oscillator/passive distinction is controlled purely by I, so the OscillatorComb designation maps directly onto the applied current without changing the model equations.

The main weakness is that FHN is phenomenological and dimensionless—voltage and time are not in physical units without rescaling. If the model needs to interface with biophysical measurements (e.g. calcium imaging timescales), a rescaling step must be added.

4.2 Izhikevich

The Izhikevich model operates in millivolts and milliseconds, making it easier to set biophysically plausible synaptic reversal potentials and time constants. The discontinuous reset rule is the main complication: it requires careful handling in stiff ODE solvers (event detection) and can produce chattering near the threshold in some parameter regimes, similar to the piecewise-linear boundary problem in the current model.

The flexibility of the four parameters is both a strength and a weakness—the parameter space is large and the relationship between parameters and firing pattern is not always intuitive.

4.3 Graded Potential

The graded model is the strongest biological choice. Non-spiking, graded-potential transmission is well-documented in *C. elegans* motor circuits [4], and most published models of *C. elegans* locomotion use this framework. It has the fewest parameters and is the least computationally expensive.

The key conceptual tradeoff is that oscillations are entirely a *network* property in this model: no single neuron oscillates in isolation. The OscillatorComb framework, which designates individual cells as oscillators, would need to be reinterpreted—oscillator cells might instead be those with higher β or I_{app}, or those in particular network positions, rather than cells with intrinsic limit-cycle dynamics.

5 Recommendation Summary

- **Preserve the current framework with minimal change:** use **FitzHugh–Nagumo**. Same parameter count, no bistability, OscillatorComb compatible.
- **Prioritise biological fidelity to *C. elegans*:** use the **graded potential model**. Fewest parameters, most appropriate neuroscience, but requires reconceptualising what an "oscillator cell" means.
- **Avoid Izhikevich** unless the spiking timescale in physical units is specifically required, as the reset rule reintroduces a variant of the threshold-crossing problem.

References

[1] Karbowski, J. & Bhatt, D. (2008). Computational model of *C. elegans* locomotion. *PLOS Computational Biology.*

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[3] Olivares, E. O. et al. (2021). Potential role of a ventral nerve cord central pattern generator in forward and backward locomotion in *C. elegans. Network Neuroscience.*

[4] Goodman, M. B. et al. (1998). Active currents regulate sensitivity and dynamic range in *C. elegans* neurons. *Neuron*, 20(4), 763–772.